



Follow-up Observations and an Improved Orbital Period for Exoplanet Candidate TOI-3505.01

Schar Scholars three month internship:
NASA Data Science Analysis
and Scholarly Research Paper
Writing

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Abstract

TOI-3505.01 is a planet candidate found by NASA's TESS mission. It circles a 6220 K star that is 1.34 times the radius of the Sun (TIC 390988385, $V = 11.2$) every 2.9151 days. On July 22, 2022 we watched the star with the 0.8 m telescope at George Mason University during the hours the transit was supposed to happen. We took 283 images in R band and kept 263 good measurements, with a scatter of 2.50 ppt. The star never dimmed (-0.66 ± 0.39 ppt). To see whether that meant anything, we added fake transits to our data. A fake 2.91 ppt transit came back at 5.7σ , so we would have caught a real one.

We saw nothing because the prediction was off. With the period we have now, the closest transit was 17.4 hours before we started, so there was nothing to catch. So we went to the TESS archive and timed 25 transits in Sectors 14, 41, 54, and 81, covering 5.05 years. Fitting a line through those times gives a period of **2.9151503 ± 0.0000057 days**, twice as precise as the TESS catalog value and matching it to within 0.4σ . It also tightens the predicted transit time for 2026 from 9 minutes to 3.

Materials and Methods

Taking and measuring the images

- Mason 0.8 m telescope, July 22, 2022 UT. R filter, 50 second exposures, $0.362''$ per pixel. We got 283 images and 281 of them lined up into one 5.07 hour run.
- We subtracted darks, divided by flats, and shifted every image so the stars matched, using 25 reference stars, then redid each step in Python to check it. Astrometry.net matched our field to sky coordinates and landed 0.47 px from our measured position.
- We measured brightness in AstrolImageJ with a **25 px ($9.1''$) aperture**, a **70–139 px sky ring**, and comparison stars C2–C11 picked from the Gaia DR3 list. We chose 25 px by trying every radius from 20 to 50 px and keeping the one with the least scatter. Our own Python photometry agrees with AstrolImageJ at $r = 0.990$.
- We cut 18 measurements for bad image quality and wrote down why for each, leaving **263**. No detrending passed the test we set up ahead of time, so the light curve is left as measured.

Working with the TESS archive

- We pulled QLP light curves for Sectors 14, 41, 54, and 81, plus the SPOC light curves and the official Data Validation reports.
- We ran a box least-squares search on each sector by itself, then fit each transit one at a time to get 25 mid-transit times and fit a straight line through them.
- We used the TESS pixel data to build difference images, find where on the sky the dimming happens, and check how much light nearby stars add. We locked in our cuts before looking at the final numbers.

Checking the neighbours

- We measured every catalog star within $60''$ on our own images across the whole night, then worked out how deep each one would have to eclipse to fake our signal once its light spills into our aperture.
- A star is cleared when our depth limit for that star sits below the depth it would need. 32 of the 44 clear this way.

Results

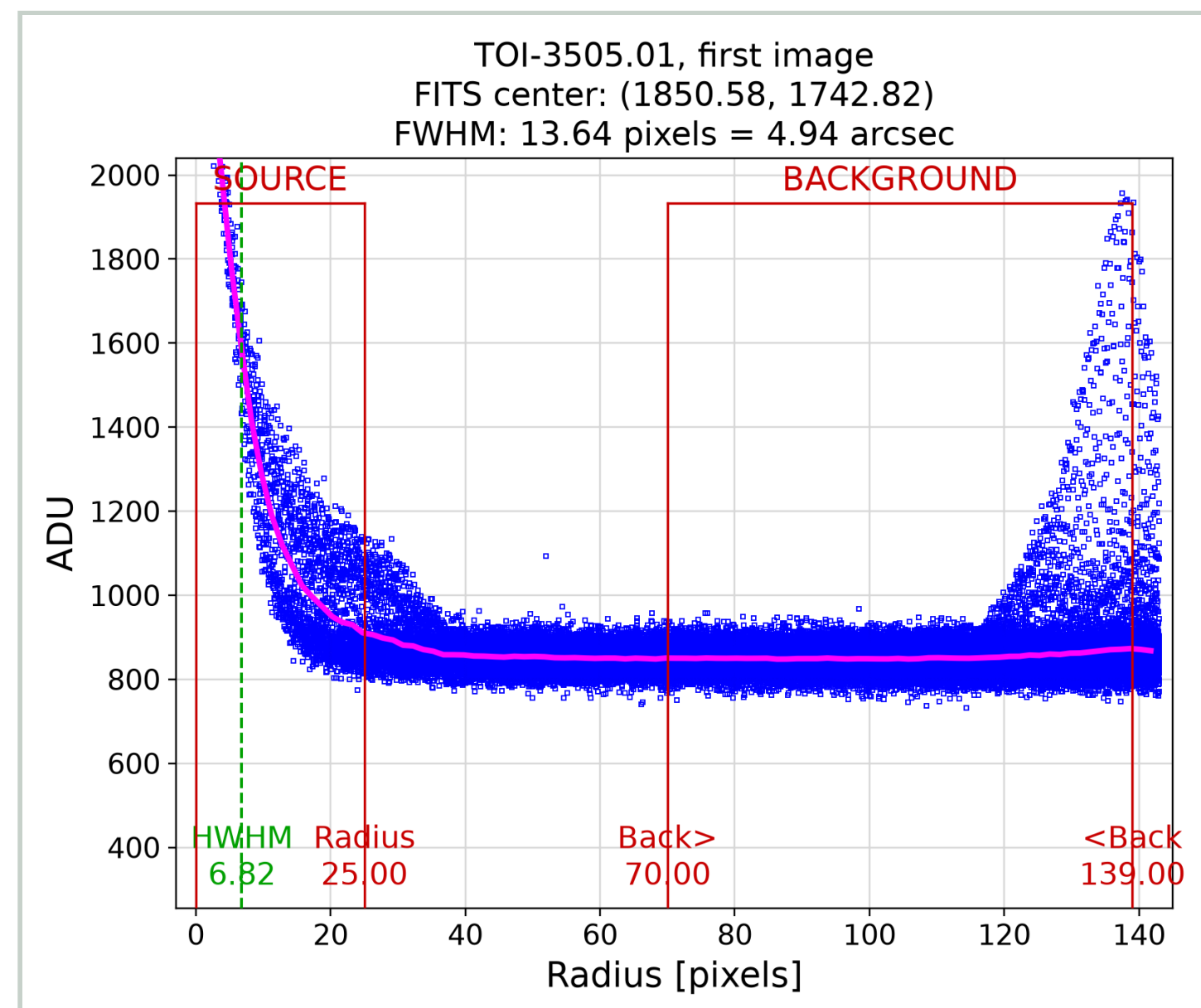


Fig. 1. Seeing profile of our star on the first image. The star's light flattens into the sky by about 25 px, which is where we set the aperture. FWHM is 13.6 px, or $4.94''$.

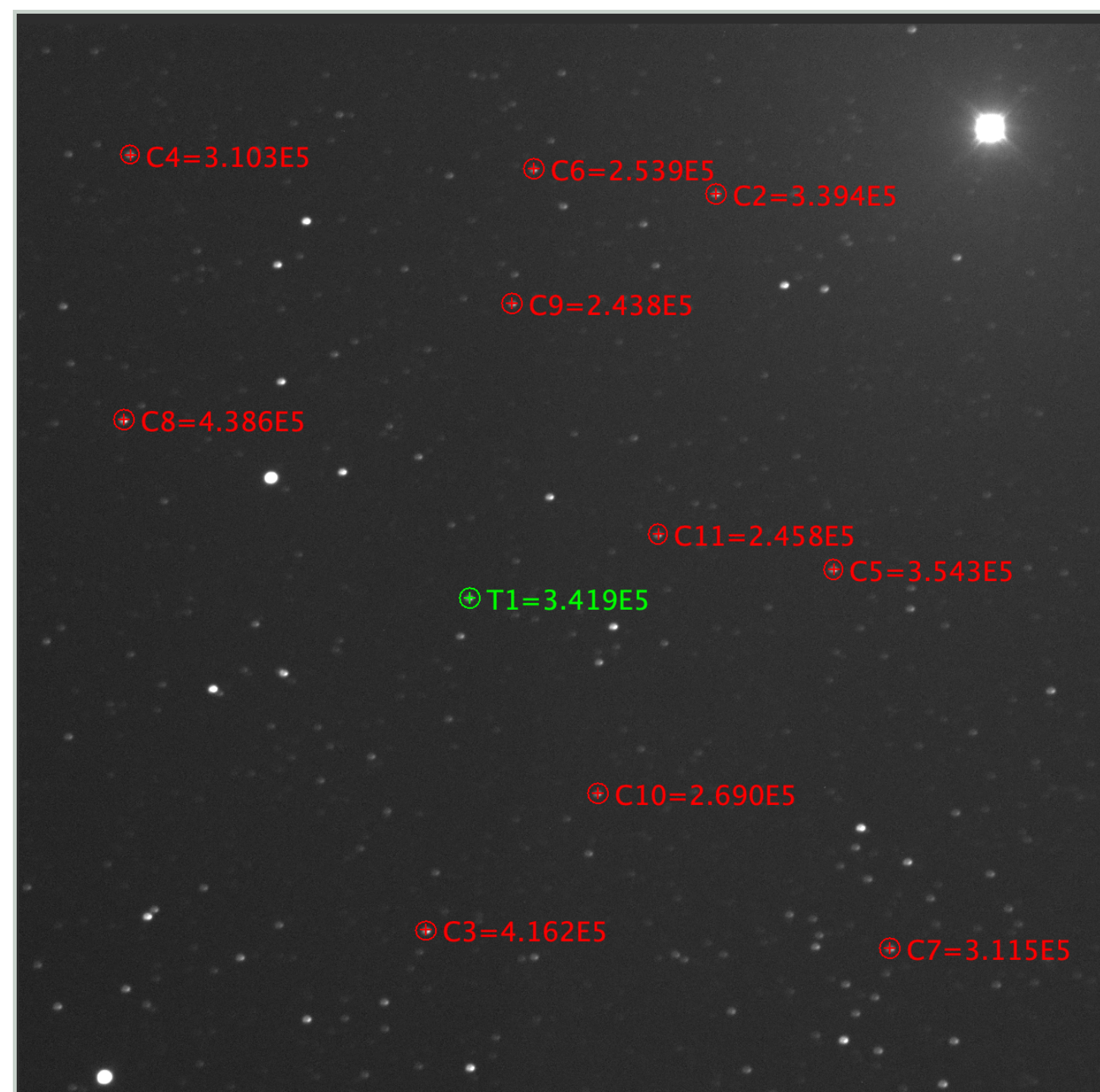


Fig. 2. Our field in AstrolImageJ. T1 is the target and C2–C11 are the comparison stars we picked off the Gaia DR3 list, with the counts each one gave.

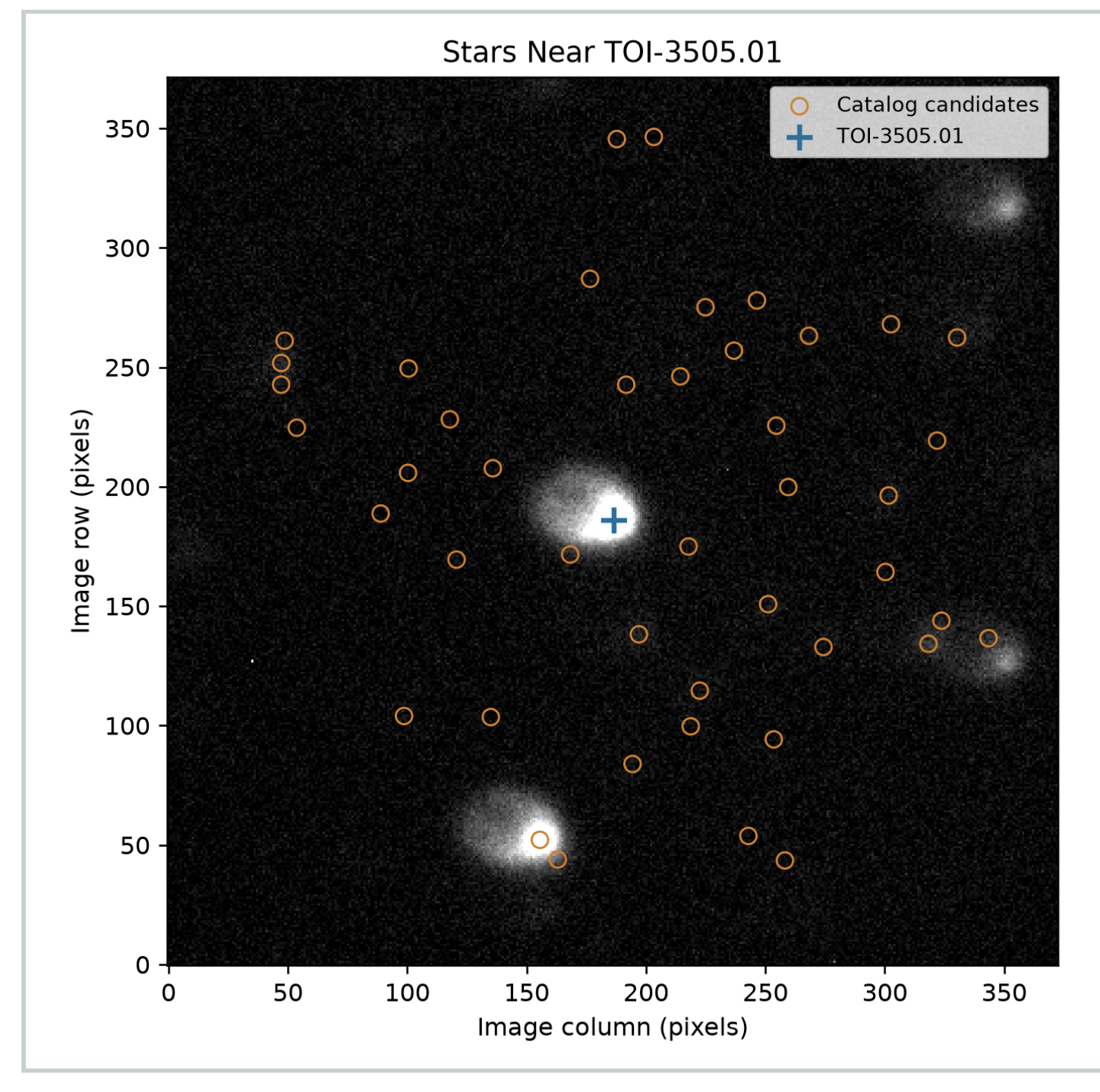


Fig. 3. The 44 catalog stars within $60''$ that we measured on our own images. Two of them fall inside our aperture, so we cannot separate those from the target.

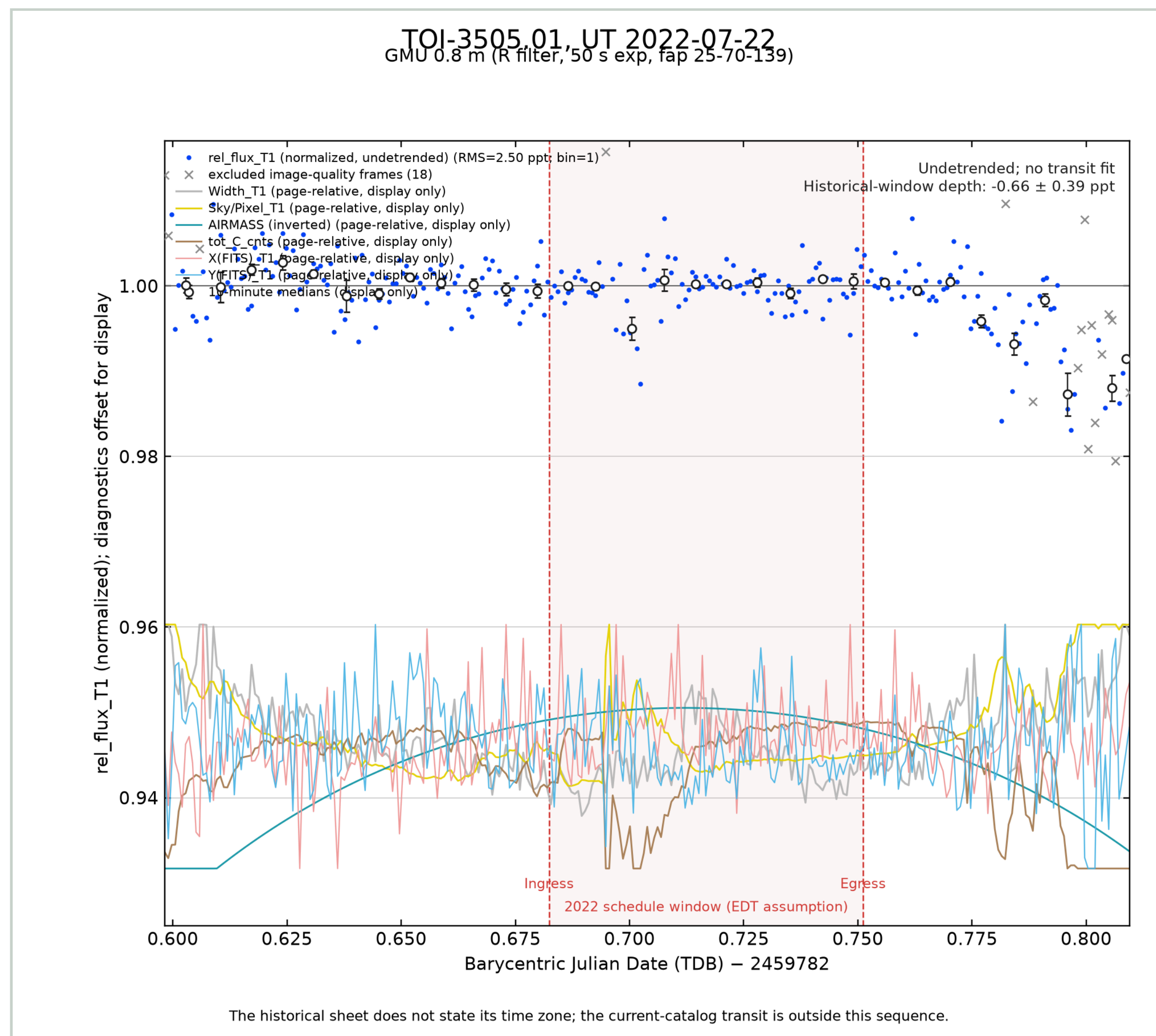


Fig. 4. Our light curve. 263 measurements at 2.50 ppt scatter. The shaded band is when the transit was scheduled and the red dashed lines are the predicted ingress and egress. Nothing dips there. The traces along the bottom are airmass, sky level, star width, and centroid position, which we checked for trends. None of them earned a detrend under our rules, so the curve is shown as measured.

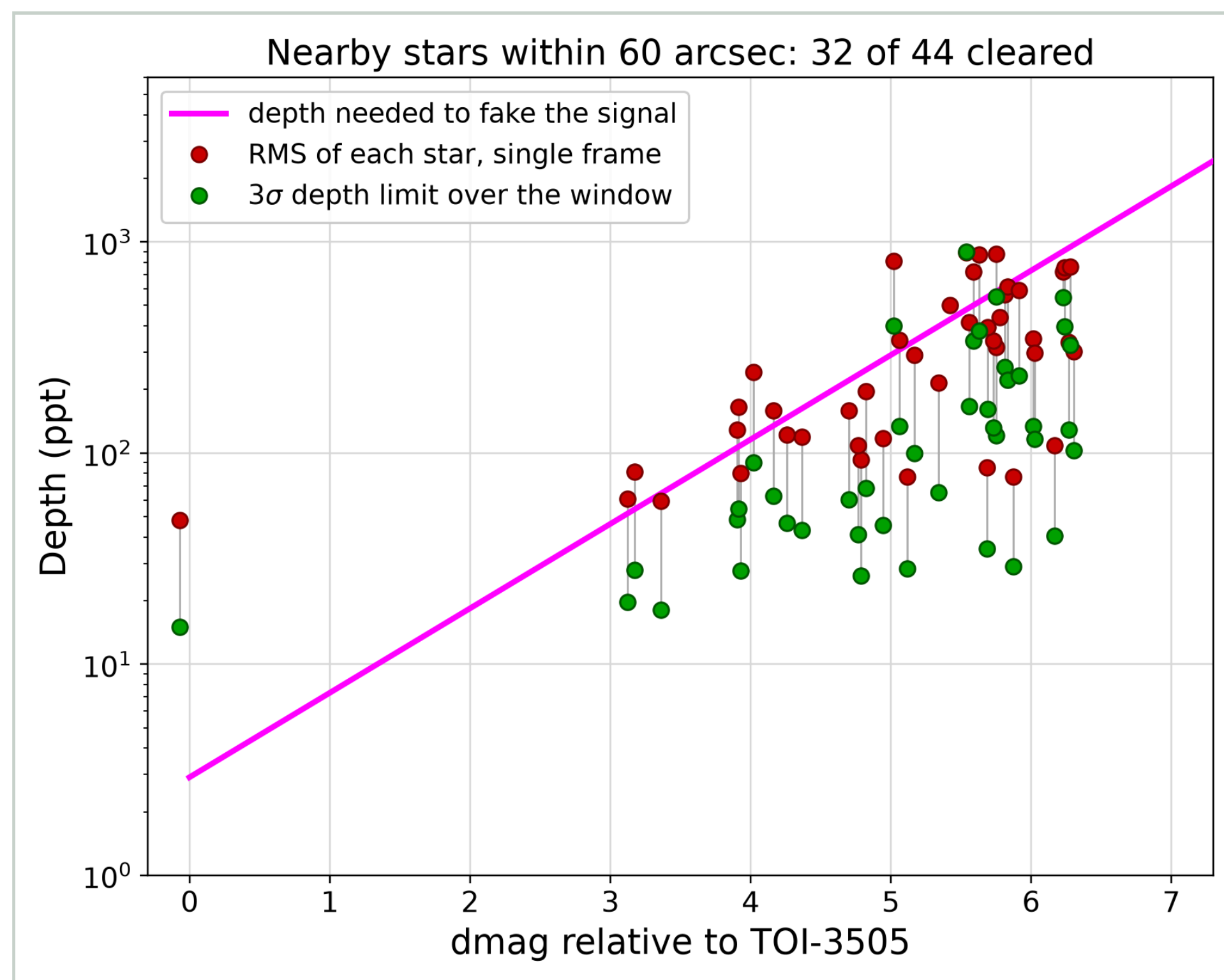


Fig. 5. Every neighbour's brightness difference against how deep we can see. The magenta line is the eclipse depth each star would need to fake our signal. Red is the scatter in a single frame, green is the depth limit after using the whole window, and the grey bars connect the same star. Green points below the line are cleared.

Discussion

Why we saw no transit

We picked this night using a 2022 prediction. With the period we have now, the nearest transit was 17.4 hours before we started, so there was nothing to catch. Our run also lands in a gap in TESS Sector 54, so we cannot line the two up. We added fake 2.91 ppt transits and pulled them back at 5.7σ , so the flat curve is a real limit, not noise.

What the TESS archive gave us

One night could not fix the prediction, so we used the archive instead. Timing 25 transits across four sectors gives **$P = 2.9151503 \pm 0.0000057$ d**. That is 2.05 times more precise than the catalog and agrees with it to 0.4σ . The timing points scatter more than their error bars allow ($\chi^2/\text{dof} = 5.3$, RMS 11.8 min). That is what we expect from Sector 14, where each image covers 30 minutes, so we widened the error bars instead of claiming the tighter number. Leaving Sector 14 out shifts the period by much less than 1σ .

Why the planet's size is uncertain

In the official SPOC fit the planet barely clips the edge of the star ($b = 0.92$). For that to work the star would have to be 3.6 times less dense than a $1.34 R_{\odot}$, $1.25 M_{\odot}$ star can be. This close to the edge, depth, length, b , and $a/R_{\star}$ all trade off. Keeping the star's density and matching our 2.06 h length gives $b = 0.85$ and $a/R_{\star} = 6.93$. Either way the planet only grazes the star, so $9.0 R_{\oplus}$ is much less certain than its error bar looks.

What we cannot claim

Another star sits $0.517''$ away and adds about 21 percent of the primary's light. Adaptive optics on the Shane 3 m sees it at $\Delta J = 1.53$ and $\Delta K_s = 1.59$, so it is redder than our target. Neither our telescope nor TESS can split the pair, so if the transit is on that star the depth and radius are bigger than we quote. Our SPOC comparisons use the same TESS data we did, so they check our method rather than add a detection. We are not the first here either: TFOP holds six other ground light curves, imaging, and five spectra, none of it published. Nobody had fit an ephemeris across all four sectors.

Checks for False Alarms

Same depth in all four sectors	$\chi^2 = 4.3 / 3$
Odd and even transits match	$\leq 1.7\sigma$
No second dip half an orbit later	$S/N \leq 1.7$
Dimming centered on our star	$1.6\text{--}5.1''$
Compared with one TESS pixel	inside $21''$
Nearby stars ruled out	32 of 44
SPOC odds of a false alarm	1×10^{-143}
Star $0.517''$ away	cannot separate

Conclusions and Future Work

- A better period.** Timing 25 transits over 5.05 years gives $P = 2.9151503 \pm 0.0000057$ d, twice as precise as the catalog value, and the 2026 transit time is now good to 3 minutes instead of 9.
- Our night still counts.** The 2.50 ppt light curve rules out a transit during the window we watched, and our fake-transit tests show we would have seen one.
- Still a good candidate.** Nothing we checked points to a false alarm. The signal looks like a roughly Jupiter-sized planet grazing the edge of its star. What limits us now is that grazing path and the star $0.517''$ away, not our photometry.
- Next:** watch a transit on the new period, since a 3 minute window makes that worth doing on a 0.8 m telescope.
- Next:** fit a real limb-darkened model with MCMC and a density prior to pin down b and the planet's size.
- Next:** pull the four-colour photometry and the six Keck velocities other groups already filed and test them against this geometry. A depth that changes with colour would mean a blended eclipsing binary.

Acknowledgements

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Major Citations

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